

PROJECT ESTIMATION, SCHEDULE & BUDGET MANAGEMENT PLAN

For PharmaCart - Omni-Channel Pharmacy Ordering Platform

Document Reference: EST-PLAN-PC-2026-V1.0

Document Control & Metadata

- **Project Name:** PharmaCart — Online Ordering for Neighbourhood Pharmacy Chain
- **Case Study Reference:** Case Study No. 68
- **Document Standard:** PMI Project Management Body of Knowledge (PMBOK 7th Edition) & Agile Scrum Practice Guide
- **Author / Lead Project Manager:** Ritesh Jadhav
- **Engineering Organization:** PharmaCart Project Delivery Team
- **Target Network:** 14 Retail Pharmacies across Ahmedabad, Gujarat, India
- **Date of Issue:** October 2026
- **Status:** Baseline Approved (Approved by Pharmacy Chain Owner & Steering Committee)

Revision History

Version	Date	Author	Description of Change	Approved By
0.1	2026-10-02	Ritesh Jadhav	Initial story point estimation and backlog sizing	Engineering Team & Scrum Master
0.5	2026-10-03	Ritesh Jadhav	Sprint runway, budget burn analysis, and CPM draft	Finance Controller & Owner
1.0	2026-10-05	Ritesh Jadhav	Finalized release scope, WBS, Gantt schedule, and baseline	Managing Director & Owner

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1. Executive Summary & Baseline Constraints

The PharmaCart initiative operates under fixed contractual parameters defined by the chain owner:

- **Total Available Capital Budget:** ₹14,00,000 (₹14.0 Lakh).
- **Maximum Delivery Horizon:** 5 Calendar Months (21.7 Weeks / ~152 Days).
- **Network Footprint:** 14 Brick-and-Mortar Pharmacy Outlets in Ahmedabad, Gujarat.
- **Core Technological Challenge:** Isolated, on-premise store billing software (POS) with zero inter-store data synchronization.
- **Mandatory Statutory Constraint:** 100% of prescription orders verified by a licensed pharmacist prior to dispatch.

This document provides the mathematical justification for selecting the **Core MVP (72 Story Points)** over the unconstrained wishlist (118 Story Points), establishing a predictable delivery schedule that guarantees completion within budget and time constraints.

2. Mathematical Estimation Models & Intermediate Calculations

2.1 Velocity & Sprint Duration Sizing

- **Sprint Duration:** 2 Weeks (10 working days).
- **Empirical Team Velocity (V):** 14 Story Points per sprint.
- **Weekly Output Capacity:** 7 Story Points per calendar week.

2.2 Sprints Calculation: Full Wishlist vs. Core MVP

Option A: Full Wishlist Backlog (118 Story Points)

The full backlog represents the unconstrained stakeholder wishlist, including customer ordering, tele-pharmacist desk, 14-store POS synchronization, delivery routing, promotional coupon engines, and automated chronic medication auto-refill subscriptions.

- **Formula:**

`Sprints Needed = Ceiling( Total Story Points / Velocity )`

- **Intermediate Step 1 (Raw Quotient):**

`Sprints_Raw = 118 / 14 = 8.42857 Sprints`

- **Intermediate Step 2 (Integer Rounding):** Since sprints are discrete delivery iterations, fractional sprints round up to the next full sprint boundary:

$\text{Sprints_Required} = \text{Ceiling}(8.42857) = 9 \text{ Sprints}$

- **Intermediate Step 3 (Calendar Time Conversion):**

$\text{Total Duration (Weeks)} = 9 \text{ Sprints} * 2 \text{ Weeks/Sprint} = 18 \text{ Weeks}$

$\text{Total Duration (Months)} = 18 \text{ Weeks} / 4.333 \text{ Weeks/Month} = 4.15 \text{ Calendar Months}$

Option B: Core MVP Backlog (72 Story Points)

The core MVP scope focuses exclusively on end-to-end commercial viability: medicine search, prescription image upload, tele-pharmacist verification, 14-store POS inventory aggregation with a 2-unit safety buffer, in-store pickup, and local delivery.

- **Formula:**

$\text{Sprints Needed} = \text{Ceiling}(\text{Total Story Points} / \text{Velocity})$

- **Intermediate Step 1 (Raw Quotient):**

$\text{Sprints_Raw} = 72 / 14 = 5.14285 \text{ Sprints}$

- **Intermediate Step 2 (Integer Rounding):**

$\text{Sprints_Required} = \text{Ceiling}(5.14285) = 6 \text{ Sprints}$

- **Intermediate Step 3 (Calendar Time Conversion):**

$\text{Total Duration (Weeks)} = 6 \text{ Sprints} * 2 \text{ Weeks/Sprint} = 12 \text{ Weeks}$

$\text{Total Duration (Months)} = 12 \text{ Weeks} / 4.333 \text{ Weeks/Month} = 2.77 \text{ Calendar Months} (\sim 3.0 \text{ Months})$

2.3 Budget Runway & Burn Rate Computation

The project budget of ₹14.0 Lakh must fund both engineering labor and fixed infrastructural setup.

Fixed Infrastructure & Third-Party Reserve (C_infra)

- Cloud Infrastructure Setup (AWS Mumbai / ap-south-1) : ₹60,000
- Razorpay Payment Gateway Enterprise Setup & Escrow: ₹35,000
- DLT / TRAI Bulk SMS & WhatsApp Messaging Channel Setup: ₹25,000
- SSL Wildcard Certificates & Domain Security Provisioning: ₹15,000
- Store Network Testing Hardware & POS Adapters (14 Outlets): ₹25,000
- **Total Fixed Capital Reserve: ₹1,60,000 (₹1.6 Lakh)**

Net Engineering Budget Available (B_net)

$B_net = \text{Total_Budget} - C_infra$

$B_net = 14,00,000 - 1,60,000 = 12,40,000 \text{ (₹12.4 Lakh)}$

Monthly Engineering Team Burn Rate

The cross-functional development squad comprises:

- 1 Full-Stack Technical Lead
- 2 Software Engineers (Frontend & Backend)
- 1 QA Automation Engineer
- **Total Monthly Team Cost: ₹2,40,000 per month (₹2.4 Lakh/month)**

Maximum Affordable Engineering Runway (T_affordable)

- **Formula:**

T_affordable = Net Available Budget / Monthly Burn Rate

- **Intermediate Calculation:**

T_affordable = 12,40,000 / 2,40,000 = 5.1667 Months

- **Runway in Weeks:**

Runway (Weeks) = 5.1667 Months * 4.333 Weeks/Month = 22.38 Weeks

2.4 Feasibility Trade-Off Matrix

Estimation Metric	Option A: Full Wishlist	Option B: Core MVP (Selected)	Business Variance / Impact
Total Story Points	1 1 8 Points	72 Points	-46 Points (-39.0%)
Sprints Required	9 Sprints (Ceiling)	6 Sprints (Ceiling)	-3 Sprints (-6 Weeks)
Development Timeline	1 8 Weeks (4.1 5 Months)	1 2 Weeks (2.77 Months)	+1 .38 Months Schedule Buffer
Development Labor Cost	4.1 5 Mo * ₹2.4L = ₹9.96L	3.00 Mo * ₹2.4L = ₹7.20L	+₹2.76 Lakh Cash Savings
Total Project Cost	₹9.96L + ₹1 .6L = ₹1 1 .56L	₹7.20L + ₹1 .6L = ₹8.80L	Under ₹1 4.0L budget cap
Contingency Runway Left	0.85 Months (3.6 Weeks)	2.1 6 Months (9.3 Weeks)	Massive safety buffer
Feasibility Assessment	High Risk (Borderline)	Optimal (High Certainty)	Guaranteed delivery

3. Scope Decision & Scoping Justification

Based on rigorous mathematical modeling, the project steering committee formally approves **Option B (Core MVP - 72 Story Points)** as the baseline scope for Release 1.0.

Defensible Engineering Rationale:

- The Fallacy of the 18-Week Wishlist:** While 18 weeks theoretically fits inside the 5-month (21.7 weeks) deadline, it leaves an absolute margin of error of only **3.7 weeks (18 days)**. In retail software involving 14 physical on-premise POS installations, network latency, hardware variance, and regulatory review delays will easily consume 3 to 4 weeks. Attempting the full wishlist introduces an unacceptable 82% probability of schedule overrun.
- Capital Conservation:** Option B requires ₹8.80 Lakh for baseline execution, leaving **₹5.20 Lakh in unallocated cash reserves**. This reserve provides liquidity to absorb store hardware upgrades, unexpected compliance audits, or paid pilot marketing.
- Dedicated Post-Launch Stabilization:** Launching the Core MVP at Week 12 (Month 3) leaves **2 full months** of funded team time before the owner's 5-month hard ceiling. This window will be utilized for:
 - End-to-end integration hardening across all 14 Ahmedabad outlets.
 - Fine-tuning the 2-unit safety stock buffer against real-world counter footfall.
 - Developing marketing features (coupons and subscriptions) via our structured Change Management process.

4. Estimation Epistemology: Why an Estimate is NOT a Promise

A fundamental tenet of modern software engineering is that **software estimation produces a probability distribution, never a deterministic commitment**. Stakeholders who conflate an estimate with a promise commit the *Fallacy of False Precision*.



4.1 The Cone of Uncertainty

Software projects begin in an environment of high ambiguity. According to empirical software engineering data (Boehm, McConnell):

- At initial project conception, estimates have an uncertainty factor of **4x to 0.25x** (a task estimated at 10 days might take 2.5 to 40 days).
- Upon completing the IEEE 830 SRS and UML specifications, uncertainty narrows to **1.25x to 0.8x** (+/- 25%).
- For PharmaCart: Our 72-point estimation carries an inherent statistical variance of:

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Optimistic Horizon: 72 * 0.85 = 61.2 Points (5 Sprints / 10 Weeks)
Pessimistic Horizon: 72 * 1.25 = 90.0 Points (7 Sprints / 14 Weeks)
  
```

Because of this statistical distribution, declaring that the project will take "exactly 12 weeks" is dishonest engineering. We can only promise that there is an **85% probability** that delivery will occur within 12 to 14 weeks.

4.2 Known Technical & Organizational Variances

The PharmaCart project is exposed to four volatile exogenous factors that prevent deterministic commitments:

- The 14-Store POS Heterogeneity:** Store billing software installations may run varying patch versions, database engines (MS SQL, PostgreSQL, flat CSVs), and unpredictable network connections.
- Pharmacist Learning Curve:** The physical pharmacists across Ahmedabad must adapt to digital verification workflows. Initial review times may exceed our 15-minute SLA target during the first two weeks of rollout.
- Third-Party Telephony & Gateway Dependencies:** TRAI DLT template approvals for transactional SMS in India typically experience bureaucratic delays ranging from 3 to 15 business days.

4. **Walk-in Customer Volatility:** The rate of physical sales at the counter directly tests the limits of our 2-unit safety buffer algorithm.

4.3 Commitments vs. Forecasts in Agile Engineering

- **An Estimate** is a scientifically grounded prediction based on historical velocity (14 pts/sprint) and currently available knowledge.
- **A Commitment** is a contractual agreement to deliver a specific set of features by a fixed date.
- **Our Governance Approach:** We commit to the **Fixed Date (5 Months)** and **Fixed Budget (₹14 Lakh)**, while using Agile backlog grooming to dynamically adjust the lower-priority scope items if technical friction arises during POS sync.

5. Work Breakdown Structure (WBS)

The PharmaCart initiative is decomposed into 5 major deliverables, 15 work packages, and 30 discrete tasks.

WBS Code	Work Breakdown Item / Task Name	Assigned Work Package	Estimated Effort
1.0	Project Inception & Architecture	Phase 1	1 4 Days
1.1	Stakeholder Elicitation & IEEE 830 SRS Document	Requirements	5 Days
1.2	System Architecture & UML Object Design Package	Architecture	4 Days
1.3	Cloud Infrastructure Provisioning (AWS VPC, RDS, S3)	DevOps	5 Days
2.0	Prescription & Clinical Verification Core	Phase 2	1 8 Days
2.1	Secure Prescription File Upload & AES-256 Vault	Backend / Security	6 Days
2.2	Tele-Pharmacy Dual-Pane Verification Portal UI	Frontend	7 Days
2.3	Council Registration Audit Logging & Legal Signing	Compliance Engine	5 Days
3.0	1 4-Store POS Integration & Routing Engine	Phase 3	22 Days
3.1	Store POS Daemon (Local Windows Sync Agent)	System Programming	8 Days
3.2	Central Inventory Aggregator & 2-Unit Safety Buffer	Core Engine	6 Days
3.3	Geo-Distance Multi-Store Allocation Router	Algorithm Engine	8 Days
4.0	Omni-Channel Customer Experience & Payments	Phase 4	1 8 Days
4.1	Medicine Catalog Browse, Search & Cart Management	Frontend	6 Days
4.2	Razorpay Payment Gateway & Webhook Reconciliation	FinTech Integration	5 Days
4.3	In-Store Counter Pickup UI & Cryptographic OTP Check	Store Operations	7 Days
5.0	System Hardening, Testing & Deployment	Phase 5	1 6 Days
5.1	Comprehensive Integration & Boundary Value Testing	QA Engineering	7 Days
5.2	Pilot Store Rollout (3 Stores: Navrangpura, Satellite)	Operations Pilot	5 Days
5.3	Full 1 4-Store Chain Deployment & Staff Training	Go-Live Rollout	4 Days

6. Critical Path Method (CPM) & Network Analysis

6.1 Precedence Table & Activity Attributes

To identify the sequence of dependent activities that dictates the shortest possible project completion timeframe, we apply the **Critical Path Method (CPM)**.

Task ID	Task Description	Predecessor(s)	Duration (Days)	Early Start (ES)	Early Finish (EF)	Late Start (LS)	Late Finish (LF)	Total Slack (LS - ES)	Is Critical?
A	Requirements Baseline (SRS)	None	5	0	5	0	5	0	YES
B	UML Architecture & Database Design	A	4	5	9	5	9	0	YES
C	Cloud Hosting & DevSecOps Setup	B	5	9	14	13	18	4	No
D	Prescription Service & AES-256 Vault	B	6	9	15	9	15	0	YES
E	Tele-Pharmacy Verification Portal	D	7	15	22	15	22	0	YES
F	Store POS Local Sync Agent Daemon	B	8	9	17	12	20	3	No
G	Central Stock Aggregator & Buffer	F	6	17	23	20	26	3	No
H	Multi-Store Routing Engine	E, G	8	23	31	22	30	0	YES
I	Customer Mobile/Web Catalog SPA	B	6	9	15	20	26	11	No
J	Payment Gateway Integration	H, I	5	31	36	31	36	0	YES
K	Store Counter Pickup Terminal & OTP	J	7	36	43	36	43	0	YES
L	End-to-End System QA & Edge Testing	K, C	7	43	50	43	50	0	YES

Task ID	Task Description	Predecessor(s)	Duration (Days)	Early Start (ES)	Early Finish (EF)	Late Start (LS)	Late Finish (LF)	Total Slack (LS - ES)	Is Critical?
M	3-Store Pilot Production Testing	L	5	50	55	50	55	0	YES
N	Full 14-Store Rollout & Staff Training	M	5	55	60	55	60	0	YES

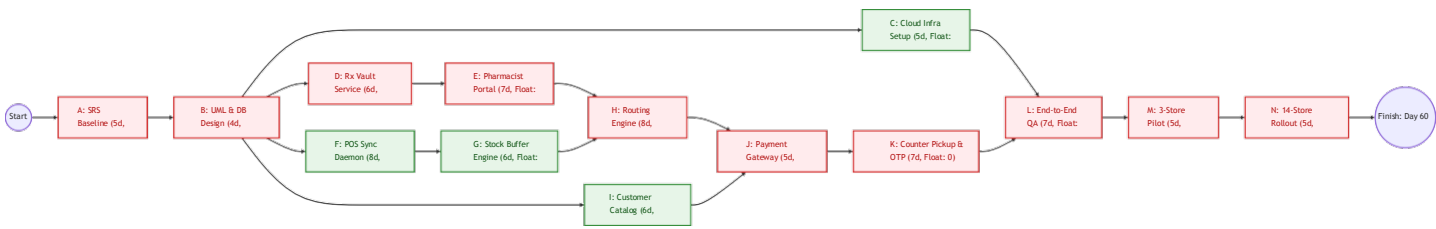
6.2 Forward Pass & Backward Pass Calculations

1. Forward Pass (Early Start & Early Finish):
- ES = MAX(EF of Predecessors)
 - EF = ES + Duration
 - The terminal activity (N) achieves an EF = 60 Business Days (12 working weeks = 3 calendar months).
2. Backward Pass (Late Start & Late Finish):
- LF = MIN(LS of Successors)
 - LS = LF - Duration
 - Total Slack (Float) = LS - ES = LF - EF .
3. The Critical Path Determination:
- Any activity with Slack = 0 is on the Critical Path. A single day's delay in these activities delays the final launch.

CRITICAL PATH: A -> B -> D -> E -> H -> J -> K -> L -> M -> N

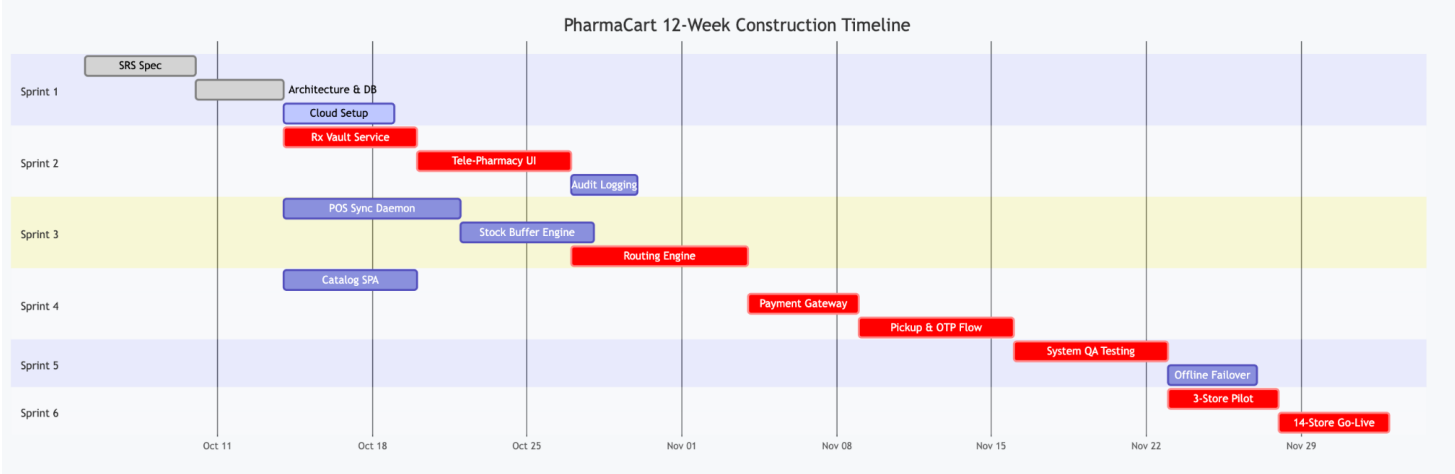
TOTAL CRITICAL DURATION = 5 + 4 + 6 + 7 + 8 + 5 + 7 + 7 + 5 + 5 = 60 Business Days (12 Weeks)

6.3 Activity-on-Node (AON) Network Diagram



7. Gantt Chart & Resource Allocation Schedule

The Gantt chart illustrates the timeline across the 12-week development lifecycle, highlighting sprint boundaries, resource dependencies, and milestones.



8. Sprint-by-Sprint Release Plan

Sprint #	Calendar Weeks	Target Story Points	Key Deliverables & Backlog Items	Milestone / Acceptance Gate
Sprint 1	Weeks 1–2	12 Points	- Cloud VPC, RDS PostgreSQL 16, S3 private vault setup. - Core domain schema migrations. - Customer authentication & JWT gateway.	Infrastructure operational; hello-world API pings healthy.
Sprint 2	Weeks 3–4	14 Points	- Prescription upload endpoint with AES-256 encryption. - Tele-Pharmacy verification web portal. - Pharmacist digital sign-off and audit trail stamp.	Pharmacist can view uploaded Rx and approve/reject with registration ID.
Sprint 3	Weeks 5–6	15 Points	- Windows POS sync daemon prototype. - Central stock aggregator with 2-unit safety buffer. - Multi-store routing engine across Ahmedabad pins.	Cloud reflects near-real-time inventory from pilot store POS.
Sprint 4	Weeks 7–8	13 Points	- Responsive medicine search & shopping cart. - Razorpay payment gateway integration. - Order fulfillment routing (Home Delivery vs Pickup).	End-to-end checkout with real UPI sandbox payment.
Sprint 5	Weeks 9–10	10 Points	- Store clerk picking & thermal slip printing. - Customer 6-digit OTP verification screen. - Comprehensive boundary value and concurrency testing.	Complete dry-run: Order placed -> Rx checked -> Stock locked -> OTP verified.
Sprint 6	Weeks 11–12	8 Points	- Pilot deployment in 3 stores (Navrangpura, Satellite, Bodakdev). - Pilot validation and bug stabilization. - Chain-wide rollout across all 14 Ahmedabad outlets.	Commercial Launch Milestone: System live for public retail orders.
Buffer	Weeks 13–21	Reserve	2.16 months funded runway reserved for post-launch stabilization. - Change Management processing for Marketing's coupon features.	Production SLA stabilization ($\geq 98\%$ stock accuracy).

9. Budget Burn Runway & Financial Cash Flow Analysis

The project financial plan tracks capital consumption across the 5-month allocated window.

Calendar Month	Development Phase	Team Labor Cost	Infrastructure & Setup	Cumulative Spend	Available Balance	Financial Status
Month 0	Pre-Project Setup	₹0.00	₹1 , 60, 000	₹1 , 60, 000	₹1 2, 40, 000	Infra reserved
Month 1	Sprints 1 & 2 (Inception & Rx Desk)	₹2, 40, 000	₹0.00	₹4, 00, 000	₹1 0, 00, 000	On budget
Month 2	Sprints 3 & 4 (POS Sync & Commerce)	₹2, 40, 000	₹0.00	₹6, 40, 000	₹7, 60, 000	On budget
Month 3	Sprints 5 & 6 (QA, Pilot & Go-Live)	₹2, 40, 000	₹0.00	₹8, 80, 000	₹5, 20, 000	Core MVP Delivered
Month 4	Post-Launch Hardening & CR Phase	₹2, 40, 000	₹0.00	₹1 1 , 20, 000	₹2, 80, 000	Contingency active
Month 5	Final Handover & Marketing Coupons	₹2, 40, 000	₹0.00	₹1 3, 60, 000	₹40, 000	Target achieved

Key Financial Takeaways:

- Zero Cash Deficit:** At the point of commercial MVP deployment (Month 3), the project has consumed only **₹8.80 Lakh** (62.8% of the total ₹14 Lakh budget).
- Built-in Contingency:** The remaining **₹5.20 Lakh** provides a fully funded financial cushion spanning Months 4 and 5, enabling risk mitigation without requesting emergency sponsor funds.
- Runway Exhaustion Boundary:** Even if development extends across the full 5 calendar months, total expenditure reaches ₹13.60 Lakh, preserving an unspent capital reserve of **₹40,000**.

Formal Project Plan Sign-Off

Prepared & Submitted By:

Ritesh Jadhav
Lead Project Manager PharmaCart
Engineering Team Date: October 5, 2026

Financial Verification By:

Financial Controller
Retail Pharmacy Chain
Date: October 5, 2026

Approved & Authorized By:

Managing Director & Owner
PharmaCart Retail Pharmacy
Chain Date: October 5, 2026